

Architecturally, any number of peripheral networks may be present in a single Fabric, including multiple networks of the same type. Nodes MAY have interfaces onto any network (hub or peripheral), and MAY communicate directly to other Nodes on the same network. However, any communication that must cross a network boundary to reach its destination SHALL flow through a Border Router. For additional details see [Section 4.2, “IPv6 Reachability”](#).

This protocol places a set of requirements on the Border Router. These requirements pertain to address assignment, route assignment and advertisement, multicast support (see [Section 4.2.1, “Stub Router Behavior”](#)), and discovery proxying (see [Section 4.3, “Discovery”](#)).

Note that in this version of the specification, Thread is the primary supported LLN. In many cases, customer installations will attempt to maintain a simple network topology, with a single Wi-Fi/Ethernet subnet, and a single Thread network. However, more than one Thread network is possible and supported.

To support home automation interoperability, this protocol supports the concept of [bridging](#) which makes available, through a data model node, devices implementing other home automation technologies, transports and link layers.

2.4. Scoped names

The Matter protocol explicitly supports multiple administrators, unrelated by any common roots of trust (multi-admin). This functionality is addressed via [multiple fabrics](#) and is enabled by the core aspects of name scoping (see below), and key considerations enabling multiple fabrics in [commissioning](#), [secure communication](#), and aspects of the data model (such as [fabric-scoped data](#)).

A Fabric is a collection of Matter devices sharing a trusted root. The root of trust in Matter is the Root CA that issues the [NOCs](#) which underpin node identities. Within the fabric, each node is uniquely identified by a stable Node ID. The scoped selection and allocation of these constructs within Matter ensures the uniqueness of identifiers and gives clear guidance on ownership and management of namespaces.

The operational root of trust—the root certificate authority (CA) as identified by its public key (Root PK)—is responsible for the allocation of correctly scoped fabric identifiers. The security of all public key infrastructures (PKI) depends on the private key of the CA being protected and neither guessable nor obtainable; that property, in turn, implies that the public key is globally unique. Within any root CA, the fabrics—identified by a 64-bit number—are unique. The uniqueness mechanism emerges from the collaboration of the commissioner and the root CA associated with that particular commissioner. Matter wraps the collaboration between the commissioner and its associated root CA and other possible data stores into a concept called the "administrative domain manager" (ADM). The algorithmic details and policies within the administrative domain manager are out of the scope of the specification as long as the allocation of all identifiers obeys the uniqueness and scoping criteria. Fabrics are uniquely identified by the tuple of their root CA's public key and a Fabric ID. Similarly, within each fabric, the administrative domain manager is responsible for assigning a unique and stable Operational Node ID to every node.

The scoping strategy as outlined here ensures that each scoped identifier can be allocated solely by the entities responsible for the scoping, without consideration for collisions or forgeries. For example, two different CAs may allocate the same fabric identifiers and this would not create any prob-